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C:\Windows\System32\cmd.exe - mysql -u root -p
Microsoft Windows [Version 10.0.22000.2176]
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C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 20
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| bookflow |
| information_schema |
| libflow  |
| mysql    |
| performance_schema |
| practice |
| sys      |
| tables   |
+-----+
8 rows in set (0.00 sec)

mysql> use practice;
Database changed
mysql> CREATE TABLE Departments (
  ->   DepartmentID INT PRIMARY KEY,
  ->   DepartmentName VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Employees (
  ->   EmployeeID INT PRIMARY KEY,
  ->   EmployeeName VARCHAR(50),
  ->   DepartmentID INT,
  ->   Salary DECIMAL(10,2),
  ->   Email VARCHAR(100),
  ->   ManagerID INT,
  ->   FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID)
  -> );
```

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Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Projects (
  ->   ProjectID INT PRIMARY KEY,
  ->   ProjectName VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Customers (
  ->   CustomerID INT PRIMARY KEY,
  ->   CustomerName VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Orders (
  ->   OrderID INT PRIMARY KEY,
  ->   CustomerID INT,
  ->   OrderDate DATE,
  ->   FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Products (
  ->   ProductID INT PRIMARY KEY,
  ->   ProductName VARCHAR(50),
  ->   Price DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Order_Details (
  ->   OrderID INT,
  ->   ProductID INT,
  ->   Quantity INT,
  ->   PRIMARY KEY (OrderID, ProductID),
  ->   FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
  ->   FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Departments VALUES
  -> (1,'CSE'),
  -> (2,'ECE'),
  -> (3,'AIDS');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Employees
  -> (EmployeeID, EmployeeName, DepartmentID, Salary, Email, ManagerID)
  -> VALUES
```

```

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-> (101,'Rahul',1,50000,'rahul@gmail.com',NULL);
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE Employees
-> SET Salary = Salary * 1.10
-> WHERE DepartmentID = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> DELETE FROM Employees
-> WHERE Salary < 30000;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Employees;
+-----+-----+-----+-----+-----+-----+
| EmployeeID | EmployeeName | DepartmentID | Salary | Email | ManagerID |
+-----+-----+-----+-----+-----+-----+
| 101 | Rahul | 1 | 55000.00 | rahul@gmail.com | NULL |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT E.EmployeeName, D.DepartmentName
-> FROM Employees E
-> INNER JOIN Departments D
-> ON E.DepartmentID = D.DepartmentID;
+-----+-----+
| EmployeeName | DepartmentName |
+-----+-----+
| Rahul | CSE |
+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT E.EmployeeName
-> FROM Employees E
-> LEFT JOIN Departments D
-> ON E.DepartmentID = D.DepartmentID
-> WHERE D.DepartmentID IS NULL;
Empty set (0.00 sec)

mysql> SELECT D.DepartmentName, E.EmployeeName
-> FROM Departments D
-> LEFT JOIN Employees E
-> ON D.DepartmentID = E.DepartmentID;
+-----+-----+
| DepartmentName | EmployeeName |
+-----+-----+
| CSE | Rahul |
| ECE | NULL |
| AIDS | NULL |
+-----+-----+

```

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+-----+-----+
3 rows in set (0.00 sec)

```
mysql> SELECT E.EmployeeName AS Employee,  
->      M.EmployeeName AS Manager  
-> FROM Employees E  
-> LEFT JOIN Employees M  
-> ON E.ManagerID = M.EmployeeID;
```

+-----+-----+
| Employee | Manager |
+-----+-----+
| Rahul | NULL |
+-----+-----+
1 row in set (0.00 sec)

```
mysql> SELECT COUNT(*) AS TotalEmployees  
-> FROM Employees;
```

+-----+
| TotalEmployees |
+-----+
| 1 |
+-----+
1 row in set (0.00 sec)

```
mysql> SELECT DepartmentID,  
->      AVG(Salary) AS AvgSalary  
-> FROM Employees  
-> GROUP BY DepartmentID;
```

+-----+-----+
| DepartmentID | AvgSalary |
+-----+-----+
| 1 | 55000.000000 |
+-----+-----+
1 row in set (0.00 sec)

```
mysql> SELECT DepartmentID,  
->      AVG(Salary) AS AvgSalary  
-> FROM Employees  
-> GROUP BY DepartmentID  
-> HAVING AVG(Salary) > 60000;  
Empty set (0.00 sec)
```

```
mysql> SELECT MAX(Salary) AS HighestSalary,  
->      MIN(Salary) AS LowestSalary  
-> FROM Employees;
```

+-----+-----+
| HighestSalary | LowestSalary |
+-----+-----+
| 55000.00 | 55000.00 |
+-----+-----+

```
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```

```
+-----+  
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO Employees VALUES  
-> (102,'Priya',1,65000,'priya@gmail.com',101),  
-> (103,'Arjun',2,70000,'arjun@gmail.com',101),  
-> (104,'Sneha',3,45000,'sneha@gmail.com',101),  
-> (106,'Riya',2,62000,'riya@gmail.com',103),  
-> (107,'Vikram',3,58000,'vikram@gmail.com',104);
```

```
Query OK, 5 rows affected (0.02 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT *  
-> FROM Employees  
-> WHERE Salary >  
-> (  
->     SELECT AVG(Salary)  
->     FROM Employees  
-> );
```

```
+-----+  
| EmployeeID | EmployeeName | DepartmentID | Salary | Email | ManagerID |  
+-----+  
| 102 | Priya | 1 | 65000.00 | priya@gmail.com | 101 |  
| 103 | Arjun | 2 | 70000.00 | arjun@gmail.com | 101 |  
| 106 | Riya | 2 | 62000.00 | riya@gmail.com | 103 |  
+-----+
```

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT *  
-> FROM Employees  
-> WHERE DepartmentID =  
-> (  
->     SELECT DepartmentID  
->     FROM Employees  
->     WHERE EmployeeName = 'Rahul'  
-> );
```

```
+-----+  
| EmployeeID | EmployeeName | DepartmentID | Salary | Email | ManagerID |  
+-----+  
| 101 | Rahul | 1 | 55000.00 | rahul@gmail.com | NULL |  
| 102 | Priya | 1 | 65000.00 | priya@gmail.com | 101 |  
+-----+
```

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT *  
-> FROM Employees  
-> WHERE Salary =  
-> (  
->     SELECT MAX(Salary)
```

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```
-> FROM Employees
-> );
```

EmployeeID	EmployeeName	DepartmentID	Salary	Email	ManagerID
103	Arjun	2	70000.00	arjun@gmail.com	101

1 row in set (0.00 sec)

```
mysql> SELECT DepartmentID,
-> COUNT(*) AS EmployeeCount
-> FROM Employees
-> GROUP BY DepartmentID
-> HAVING COUNT(*) > 5;
Empty set (0.00 sec)
```

```
mysql> SELECT Salary
-> FROM
-> (
-> SELECT Salary,
-> DENSE_RANK() OVER (ORDER BY Salary DESC) AS Rnk
-> FROM Employees
-> ) X
-> WHERE Rnk = 2;
```

Salary
65000.00

1 row in set (0.01 sec)

```
mysql> SELECT *
-> FROM Employees E
-> WHERE Salary >
-> (
-> SELECT AVG(Salary)
-> FROM Employees
-> WHERE DepartmentID = E.DepartmentID
-> );
```

EmployeeID	EmployeeName	DepartmentID	Salary	Email	ManagerID
102	Priya	1	65000.00	priya@gmail.com	101
103	Arjun	2	70000.00	arjun@gmail.com	101
107	Vikram	3	58000.00	vikram@gmail.com	104

3 rows in set (0.00 sec)

```
mysql> SELECT C.CustomerID,
```

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```
-> C.CustomerName
-> FROM Customers C
-> LEFT JOIN Orders O
-> ON C.CustomerID = O.CustomerID
-> WHERE O.OrderID IS NULL;
```

Empty set (0.00 sec)

```
mysql> SELECT P.ProductID,
-> P.ProductName,
-> SUM(OD.Quantity) AS TotalSold
-> FROM Products P
-> JOIN Order_Details OD
-> ON P.ProductID = OD.ProductID
-> GROUP BY P.ProductID, P.ProductName
-> ORDER BY TotalSold DESC
-> LIMIT 1;
```

Empty set (0.00 sec)

```
mysql> SELECT DepartmentID,
-> COUNT(*) AS EmployeeCount
-> FROM Employees
-> GROUP BY DepartmentID
-> HAVING COUNT(*) =
-> (
-> SELECT MAX(EmpCount)
-> FROM
-> (
-> SELECT COUNT(*) AS EmpCount
-> FROM Employees
-> GROUP BY DepartmentID
-> ) X
-> );
```

DepartmentID	EmployeeCount
1	2
2	2
3	2

3 rows in set (0.00 sec)

```
mysql> SELECT
-> YEAR(OrderDate) AS Year,
-> MONTH(OrderDate) AS Month,
-> COUNT(*) AS TotalOrders,
-> SUM(COUNT(*)) OVER
-> (
-> ORDER BY YEAR(OrderDate),
-> MONTH(OrderDate)
```

```

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->         MONTH(OrderDate)
->     ) AS RunningTotal
-> FROM Orders
-> GROUP BY YEAR(OrderDate),
->         MONTH(OrderDate);
Empty set (0.00 sec)

mysql> SELECT
->     C.CustomerID,
->     C.CustomerName,
->     SUM(P.Price * OD.Quantity) AS TotalPurchase,
->     RANK() OVER
->     (
->         ORDER BY SUM(P.Price * OD.Quantity) DESC
->     ) AS CustomerRank
-> FROM Customers C
-> JOIN Orders O
-> ON C.CustomerID = O.CustomerID
-> JOIN Order_Details OD
-> ON O.OrderID = OD.OrderID
-> JOIN Products P
-> ON OD.ProductID = P.ProductID
-> GROUP BY C.CustomerID,
->         C.CustomerName;
Empty set (0.00 sec)

mysql> SELECT E.EmployeeName,
->     E.Salary,
->     M.EmployeeName AS Manager,
->     M.Salary AS ManagerSalary
-> FROM Employees E
-> JOIN Employees M
-> ON E.ManagerID = M.EmployeeID
-> WHERE E.Salary > M.Salary;
+-----+-----+-----+-----+
| EmployeeName | Salary | Manager | ManagerSalary |
+-----+-----+-----+-----+
| Priya        | 65000.00 | Rahul   | 55000.00 |
| Arjun        | 70000.00 | Rahul   | 55000.00 |
| Vikram       | 58000.00 | Sneha   | 45000.00 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT Email,
->     COUNT(*) AS EmailCount
-> FROM Employees
-> GROUP BY Email
-> HAVING COUNT(*) > 1;
Empty set (0.00 sec)

```